Understanding the Inner Workings of Language Models Through Representation Dissimilarity

Davis Brown[1] **Charles Godfrey**[2,†] **Nicholas Konz**[1,3] **Jonathan Tu**[1] **Henry Kvinge**[1,4]
[1]Pacific Northwest National Laboratory [2]Thomson Reuters Labs
[3]Duke University [4]University of Washington
{first.last}@pnnl.gov

arXiv:2310.14993v1 [cs.LG] 23 Oct 2023

Abstract

As language models are applied to an increasing number of real-world applications, understanding their inner workings has become an important issue in model trust, interpretability, and transparency. In this work we show that representation dissimilarity measures, which are functions that measure the extent to which two model's internal representations differ, can be a valuable tool for gaining insight into the mechanics of language models. Among our insights are: (i) an apparent asymmetry in the internal representations of model using SoLU and GeLU activation functions, (ii) evidence that dissimilarity measures can identify and locate generalization properties of models that are invisible via in-distribution test set performance, and (iii) new evaluations of how language model features vary as width and depth are increased. Our results suggest that dissimilarity measures are a promising set of tools for shedding light on the inner workings of language models.

1 Introduction

The defining feature of deep neural networks is their capability of learning useful feature representations from data in an end-to-end fashion. Perhaps ironically, one of the most pressing scientific challenges in deep learning is *understanding* the features that these models learn. This challenge is not merely philosophical: learned features are known to impact model interpretability/explainability, generalization, and transferability to downstream tasks, and it can be the case that none of these effects are visible from the point of view of model performance on even carefully selected validation sets.

When studying hidden representations in deep learning, a fundamental question is whether the internal representations of a given pair of models are similar or not. Dissimilarity measures (Klabunde et al., 2023) are a class of functions that seek to address this by measuring the difference between (potentially high-dimensional) representations. In this paper, we focus on two such functions that, while popular in computer vision, have seen limited application to language models: model stitching (Lenc and Vedaldi, 2014; Bansal et al., 2021) and centered kernel alignment (CKA) (Kornblith et al., 2019). *Model stitching* extracts features from earlier layers of model f and plugs them into the later layers of model g (possibly mediated by a small, learnable, connecting layer), and evaluates downstream performance of the resulting "stitched" model. Stitching takes a task-centric view towards representations, operating under the assumption that if two models have similar representations then these representations should be reconcilable by a simple transformation to such an extent that the downstream task can still be solved. On the other hand, CKA compares the statistical structure of the representations obtained in two different models/layers from a fixed set of input datapoints, ignoring any relationship to performance on the task for which the models were trained.

In this paper we make the case that dissimilarity measures are a tool that has been underutilized in the study of language models. We support this claim through experiments that shed light on the inner workings of language models: **(i)** We show that stitching can be used to better understand the changes to representations that result from using different nonlinear activations in a model. In particular, we find evidence that feeding Gaussian error linear unit (GeLU) model representations into a softmax linear unit (SoLU) model incurs a smaller penalty in loss compared to feeding SoLU activations into a GeLU model, suggesting that models with GeLU activations may form representations that contain strictly more useful information for the training task than the representations of models using SoLU activations. **(ii)** We show that dissimilarity measures can localize differences in models

† Work done at Pacific Northwest National Laboratory.

that are invisible via test set performance. Following the experimental set-up of (Juneja et al., 2022), we show that both stitching and CKA detect the difference between models which generalize to an out-of-distribution test set and models that do not generalize. **(iii)** Finally, we apply CKA to the Pythia networks (Biderman et al., 2023), a sequence of generative transformers of increasing width and depth, finding a high degree of similarity between early layer features even as scale varies from 70 million to 1 billion parameters, an emergent "block structure" previously observed in CKA analysis of image classifiers such as ResNets and Vision Transformers, and showing that CKA identifies a Pythia model (`pythia-2.8b-deduped`) exhibiting remarkably low levels of feature similarity when compared with the remaining models in the family (as well as inconsistent architectural characteristics). This last finding is perhaps surprising considering the consistent trend towards lower perplixity and higher performance on benchmarks with increasing model scale seen in the original Pythia evaluations (Biderman et al., 2023) — one might have expected to see an underlying consistent evolution of hidden features from the perspective of CKA.

2 Background

In this section we review the two model dissimilarity measures appearing in this paper.

Model stitching (Bansal et al., 2021): Informally, model stitching asks how well the representation extracted by the early layers of one model can be used by the later layers of another model to solve a specific task. Let f be a neural network and for a layer l of f let $f_{\leq l}$ (respectively $f_{\geq l}$) be the composition of the first l layers of f (respectively the layers m of f with $m \geq l$). Given another network g the model obtained by stitching layer l of f to layer m of g with stitching layer φ is $g_{>m} \circ \varphi \circ f_{\leq l}$. The performance of this stitched network measures the similarity of representations of f at layer l and representations of g at layer m.

Centered kernel alignment (CKA) (Kornblith et al., 2019): Let $D = \{x_1, \ldots, x_d\}$ be a set of model inputs. For models f and g with layers l and m respectively, let $A_{f,l,g,m}$ be the covariance matrix of $f_{\leq l}(D)$ and $g_{\leq m}(D)$. Then the CKA score for models f and g at layers l and m respectively and evaluated at D is

$$\frac{||A_{f,l,g,m}||_F^2}{||A_{f,l,f,l}||_F ||A_{g,m,g,m}||_F}, \tag{1}$$

where $||\cdot||_F$ is the Frobenious norm. Higher CKA scores indicate more structural similarity between representations. In our experiments we use an unbiased estimator of eq. (1) to calculate CKA in batches (see appendix C for details).

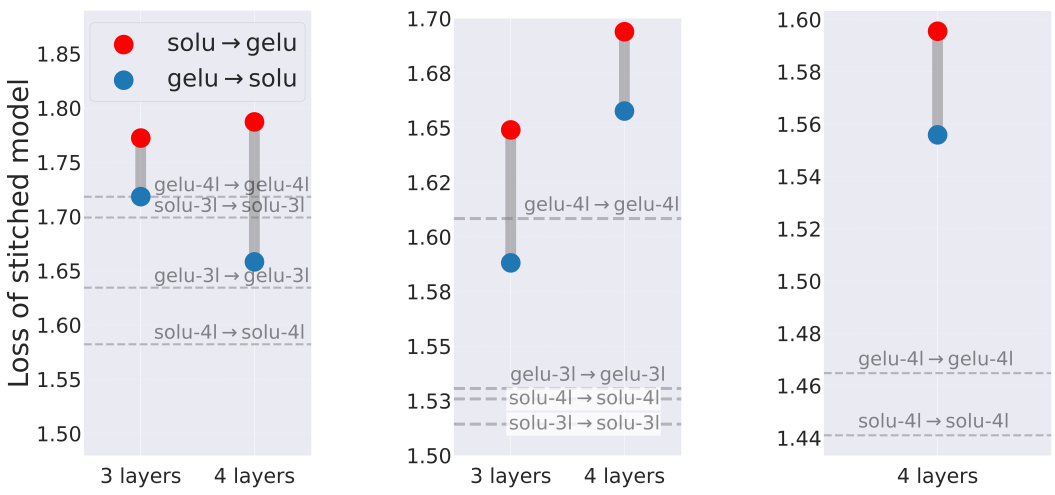
3 Model stitching reveals an asymmetry between GeLU and interpretable-by-design SoLU

The design of more interpretable models is an area of active research. Interpretable-by-design models often achieve comparable performance on downstream tasks to their non-interpretable counterparts (Rudin, 2019). However, these models (almost by definition) have differences in their hidden layer representations that can impact downstream performance.

Many contemporary transformers use the Gaussian error linear unit activation function, approximately $\mathrm{GeLU}(x) = x * \mathrm{sigmoid}(1.7x)$. The softmax linear unit $\mathrm{SoLU}(x) = x \cdot \mathrm{softmax}(x)$ is an activation function introduced in (Elhage et al., 2022) in an attempt to reduce neuron polysemanticity: the softmax has the effect of shrinking small and amplifying large neuron outputs. One of the findings of (Elhage et al., 2022) was that SoLU transformers yield comparable performance to their GeLU counterparts on a range of downstream tasks. However, those tests involved zero-shot evaluation or fine-tuning of the full transformer, and as such they do not shed much light on intermediate hidden feature representations. We use stitching to do this.

Using the notation from Section 2, we set our stitching layer φ to be a learnable linear layer (all other parameters of the stitched model are frozen) between the residual streams following a layer l. We stitch small 3-layer (9.4M parameter) and 4-layer (13M parameter) SoLU and GeLU models trained by (Nanda, 2022), using the Pile validation set (Gao et al., 2020) to optimize φ.

Figure 1 displays the resulting stitching losses calculated on the Pile validation set. For both the 3-layer and 4-layer models, and at every layer considered, we see that when f is a SoLU model for the stitched model $g_{>m} \circ \varphi \circ f_{\leq l}$ — the "SoLU-into-GeLU" cases — larger penalties are incurred than when f uses GeLUs, i.e. the "GeLU-into-SoLU"



(a) Stitching at layer 1 (b) Stitching at layer 2 (c) Stitching at layer 3

Figure 1: **Comparing stitching loss** between 3-layer and 4-layer GeLU and SoLU models on the Pile validation set. Stitching for "solu→gelu," i.e. stitching with a SoLU 'head' and GeLU 'tail,' incurs systematic penalties (larger loss is worse) compared to the "gelu→solu" stitching. Baselines for stitching identical models (e.g., "gelu-3l→gelu-3l") are given to account for potential inherent differences in learning the stitching layer φ between the activation functions.

cases. We conjecture that this outcome results from the fact that SoLU activations effectively reduce capacity of hidden feature layers; there may be an analogy between the experiments of fig. 1 and those of (Bansal et al., 2021, Fig. 3c) stitching vision models of different widths.

We also measure the stitching performance between pairs of identical models, the SoLU-into-SoLU and GeLU-into-GeLU stitching baselines. This is meant to delineate between two factors influencing the stitching penalties being recorded: the ease of optimizing stitching layers and the interplay between hidden features of possibly different architectures. We seek to measure differences between hidden features, while the former is an additional unavoidable factor inherent in model stitching experiments. The identical SoLU-into-SoLU and GeLU-into-GeLU stitching baselines serve as a proxy measure of stitching optimization success, since in-principle the stitching layer should be able to learn the identity matrix and incur a stitching penalty of 0. So, that SoLU-into-SoLU stitches better than GeLU-into-GeLU gives evidence for SoLU models being easier to stitch with than GeLU models, from an optimization perspective. That both SoLU-into-GeLU and GeLU-into-SoLU incur higher stitching penalties than GeLU-into-GeLU suggests that the penalties cannot only be a result of stitching layer optimization issues.

As pointed out in (Bansal et al., 2021) such an analysis is not possible using CKA, which only detects distance between distributions of hidden features (displayed for our GeLU and SoLU models in figs. 4 and 5), not their usefulness for a machine learning task. Further, the differences in layer expressiveness between GeLU and SoLU models are not easily elicited by evaluation on downstream tasks, but can be seen through linear stitching.

4 Locating generalization failures

Starting with a single pretrained BERT model and fine-tuning 100 models (differing only in random initialization of their classifier heads and stochasticity of batching in fine-tuning optimization) on the Multi-Genre Natural Language Inference (MNLI) dataset (Williams et al., 2018), the authors of (McCoy et al., 2020) observed a striking phenomenon: despite the fine-tuned BERTs' near identical performance on MNLI, their performance on Heuristic Analysis for NLI Systems (HANS), an out of distribution variant of MNLI, (McCoy et al., 2019) was highly variable. On a specific subset of HANS called "lexical overlap" (HANS-LO) one subset of fine-tuned BERTs (the *generalizing*) models) were relatively successful and used syntactic features; models in its complement (the *heuristic* models) failed catastrophically and used a strategy akin to